

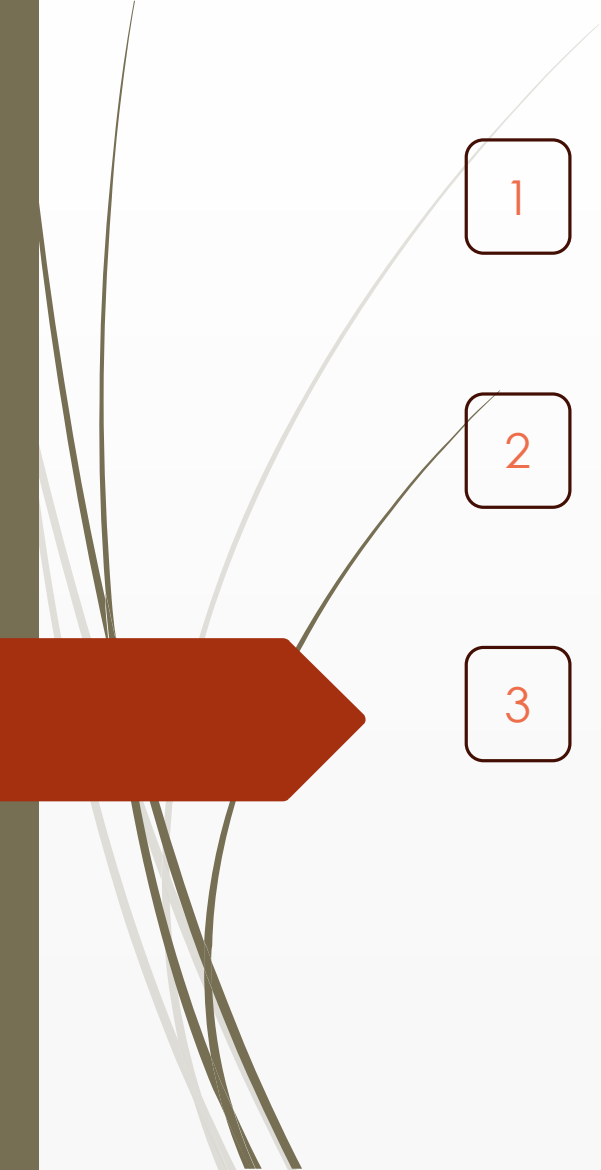


Provide Insights to Management in Consumer Goods Domain

AD-HOC REQUESTS

Resume Challenge – 4 – SQL

CONTENTS:



1

Company Overview

2

Challenges

3

AD-Hoc requests & Insights

Company overview:

Atliq Hardwares (imaginary company) is one of the leading computer hardware producers in India and well expanded in other countries too. They produce Mouse, Keyboard , PC and so on.

The company's fiscal year starts from September to August.



CHALLENGE:

Management faced challenges as they couldn't access insights needed to make smarter, data-driven decisions for their business.

To solve this issue, their Analytics Director decided to hire several Junior Data Analysts and assigned them the task of handling ad-hoc requests and providing valuable insights.



Requests - 1

Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

Query

```
SELECT DISTINCT MARKET  
FROM DIM_CUSTOMER  
WHERE CUSTOMER = 'ATLIQ EXCLUSIVE' AND REGION = 'APAC';
```

Output

MARKET
India
Indonesia
Japan
Philippines
South Korea
Australia
Newzealand
Bangladesh



AtliQ Exclusive operates in **8** countries in APAC region.

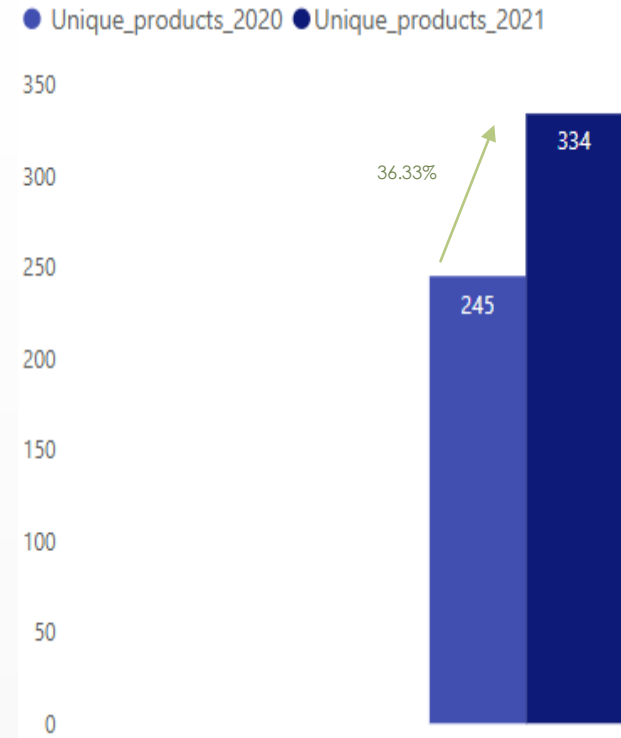
What is the percentage of unique product increase in 2021 vs. 2020?

Query

```
with products as (  
  select  
  
    count(distinct(case when fiscal_year =2020 then product_code end)) as unique_product_2020,  
    count(distinct(case when fiscal_year =2021 then product_code end)) as unique_product_2021  
  from fact_sales_monthly  
)  
select  
  unique_product_2020,  
  unique_product_2021,  
  CONCAT(ROUND(((unique_product_2021-unique_product_2020)/unique_product_2020)*100,2),"%") as percentage_chg  
from products;
```

Output

unique_products_2020	unique_products_2021	percentage_chg
245	334	36.33%



The total percentage of products increased from 2020 to 2021 is **36.33%**.

Requests - 3

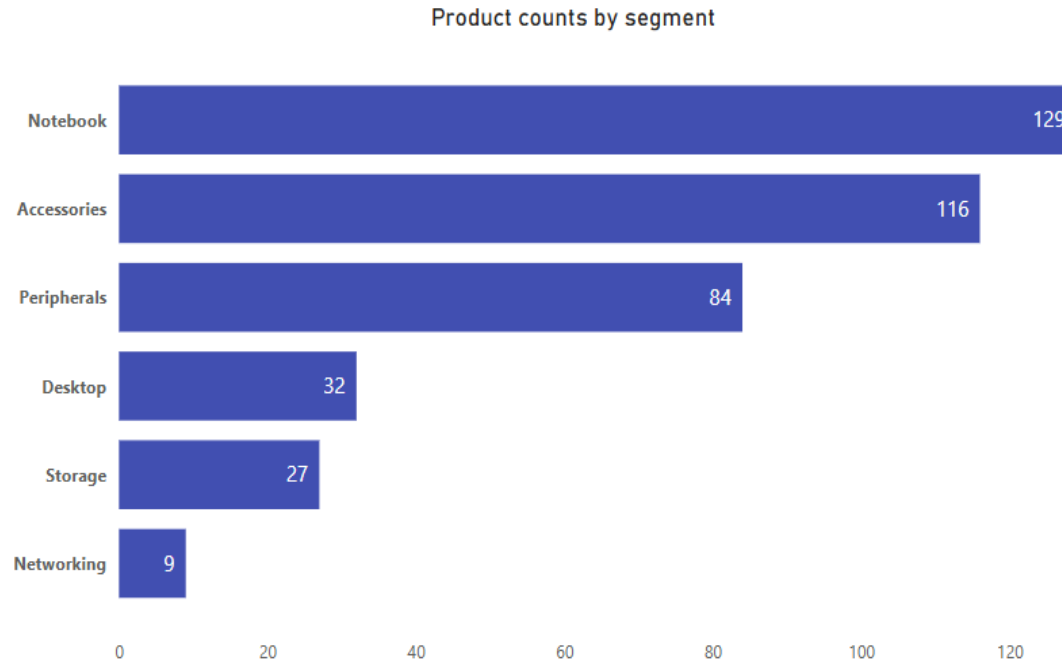
Provide a report with all the unique product counts for each segment and sort them in descending order of product counts.

Query

```
select
  segment,
  count(distinct (product_code)) as product_counts
from dim_product
group by segment
order by product_counts desc;
```

Output

segment	product_counts
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9



In the segments **Notebook, Accessories, and Peripherals** total account for **82.87%**, which is the highest.

In contrast, **Desktop, Storage, and Networking** have the lowest product count at **17.13%**.

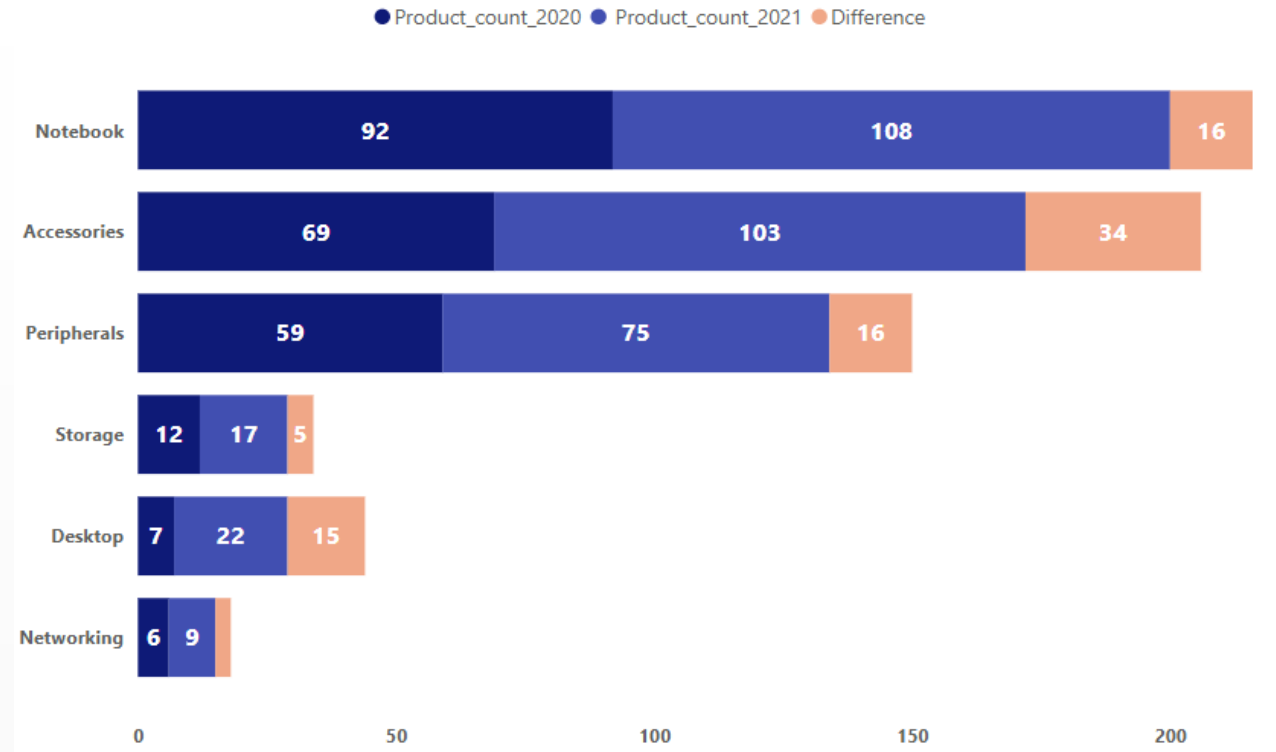
Follow-up: Which segment had the most increase in unique products in 2021 vs 2020?

Query

```
with cte as (  
  select  
    p.segment as segment,  
    count(distinct(case when s.fiscal_year = 2020 then p.product_code end)) as product_count_2020,  
    count(distinct(case when s.fiscal_year = 2021 then p.product_code end)) as product_count_2021  
  from fact_sales_monthly as s  
  join dim_product as p  
  on s.product_code = p.product_code  
  group by p.segment  
)  
select  
  segment,  
  product_count_2020,  
  product_count_2021,  
  ( product_count_2021 - product_count_2020) as difference  
from cte  
order by difference desc;
```

Output

segment	product_count_2020	product_count_2021	difference
Accessories	69	103	34
Notebook	92	108	16
Peripherals	59	75	16
Desktop	7	22	15
Storage	12	17	5
Networking	6	9	3



The production of **Accessories** segment increased from 2020 to 2021 by a difference of 34.

Get the products that have the highest and lowest manufacturing costs.

Query

```
select
    mc.product_code,
    p.product,
    round(mc.manufacturing_cost,2) as manufacturing_cost
from fact_manufacturing_cost mc
join dim_product p
on p.product_code = mc.product_code
where mc.manufacturing_cost = (select max(manufacturing_cost) from fact_manufacturing_cost) or
      mc.manufacturing_cost =(select min(manufacturing_cost) from fact_manufacturing_cost)
order by mc.manufacturing_cost desc;
```

Output

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.54
A2118150101	AQ Master wired x1 Ms	0.89

\$240.54

AQ HOME Allin 1 Gen 2

\$0.89

AQ Master wired x1 Ms

The highest manufacturing costs is AQ Home Allin1 Gen 2 (Desktop) by **\$240.54/-** and the lowest manufacturing costs is AQ Master Wired x1 Ms (Mouse) is **\$0.89/-** .

Requests - 6

Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market.

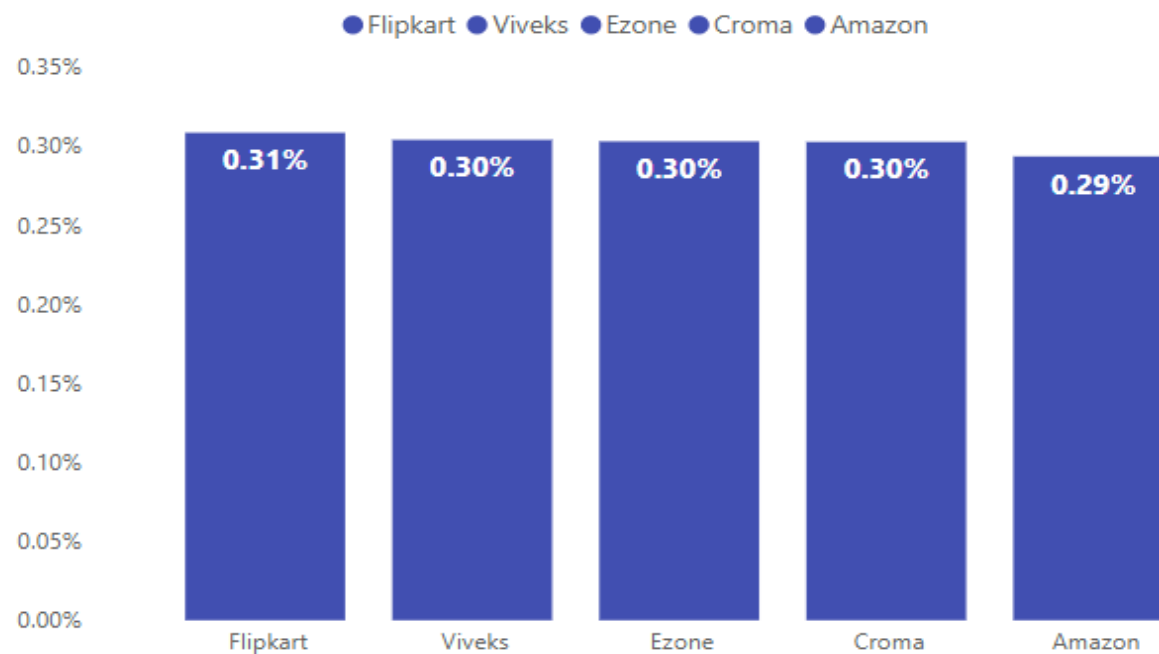
Query

```
select
    fp.customer_code,
    c.customer,
    concat(round(avg(fp.pre_invoice_discount_pct),2),'%') as average_discount_percentage
from fact_pre_invoice_deductions fp
join dim_customer c
on c.customer_code= fp.customer_code
Where fiscal_year = 2021 and c.market = 'india'
group by c.customer_code ,fp.customer_code
order by average_discount_percentage desc
LIMIT 5;
```

Output

customer_code	customer	average_discount_percentage
90002009	Flipkart	0.31%
90002006	Viveks	0.30%
90002002	Croma	0.30%
90002003	Ezone	0.30%
90002016	Amazon	0.29%

Top 5 customers by avg_discount_percentage



In 2021, Out of top 5 customers Flipkart had the highest average pre invoice discount of **30.83%**, while Amazon recorded at the lowest at **29.33%**.

Requests - 7

Get the complete report of the Gross sales amount for the customer "Atliq Exclusive" for each month .

This analysis helps to get an idea of low and high-performing months and take strategic decisions.

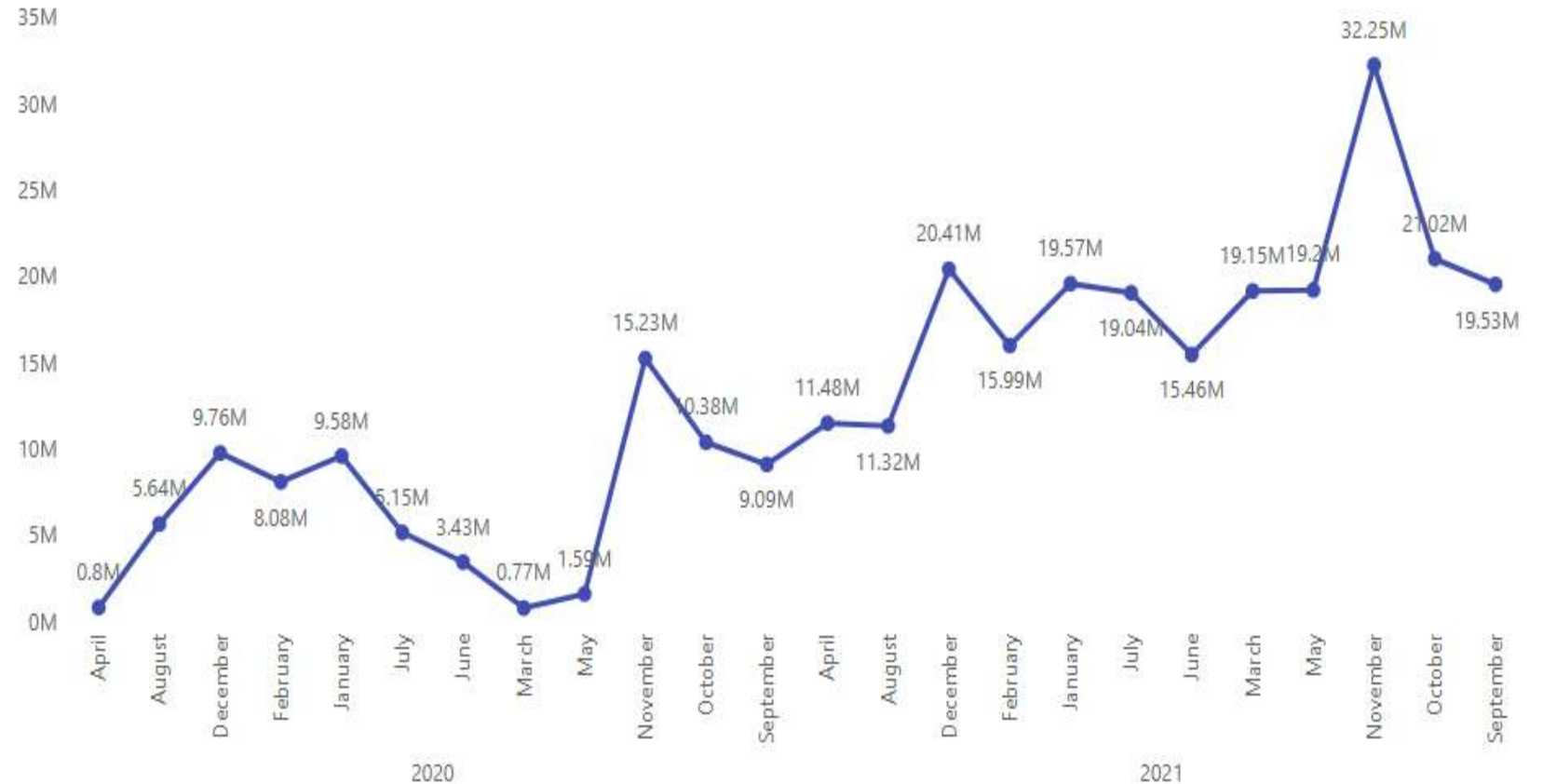
Query

```
select
    Monthname(s.date) as Month,
    s.fiscal_year as Year,
    concat(round(sum(s.sold_quantity*g.gross_price)/1000000,2),"M") as Gross_sales_Amount
From fact_sales_monthly s
join fact_gross_price g
on g.product_code = s.product_code
join dim_customer c
on c.customer_code = s.customer_code
where customer = 'Atliq Exclusive'
group by Month, Year;
```

Output

Month	Year	Gross_sales_Amount
September	2020	9.09M
November	2020	15.23M
December	2020	9.76M
January	2020	9.58M
March	2020	0.77M
April	2020	0.80M
May	2020	1.59M
July	2020	5.15M
August	2020	5.64M
September	2021	19.53M
November	2021	32.25M
December	2021	20.41M
January	2021	19.57M
March	2021	19.15M
April	2021	11.48M
May	2021	19.20M
July	2021	19.04M
August	2021	11.32M
October	2020	10.38M
February	2020	8.08M
June	2020	3.43M
October	2021	21.02M
February	2021	15.99M
June	2021	15.46M

Gross_sales_Amount by Year and Month



Atliq recorded its lowest sales in the month of **March-2020** i.e.**0.77M** due to the pandemic. While it surged in the **November-2021** month i.e.**32.25M**.

In which quarter of 2020, got the maximum total_sold_quantity?

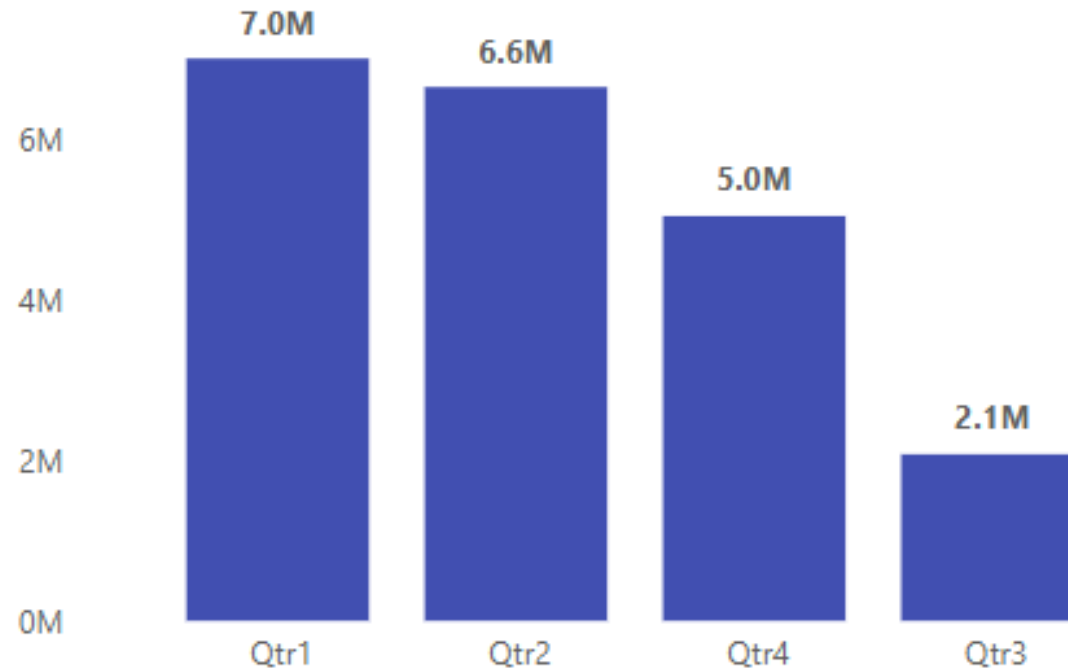
Query

```
SELECT
CASE
    WHEN MONTH(date) in ( 9,10, 11) THEN ' Qtr1'
    WHEN MONTH(date) in(12,1,2) THEN 'Qtr2'
    WHEN MONTH(date) in (3,4,5) THEN 'Qtr3'
    WHEN MONTH(date) in (6,7,8) THEN 'Qtr4'
END AS quarter,
sum(sold_quantity) as total_sold_quantity
from fact_sales_monthly
where fiscal_year = 2020
group by Quarter
order by total_sold_quantity desc ;
```

Output

Quarter	Total_sold_quantity
Qtr1	7005619
Qtr2	6649642
Qtr4	5042541
Qtr3	2075087

Total_sold_quantity by Quarter



Quarter 1 saw the highest sales at **7.0M units**, while **Quarter 3** dropped to **2.1M**. Sales rebounded in **Quarter 4**, increasing by **3.0M units** to reach **5.0M**.

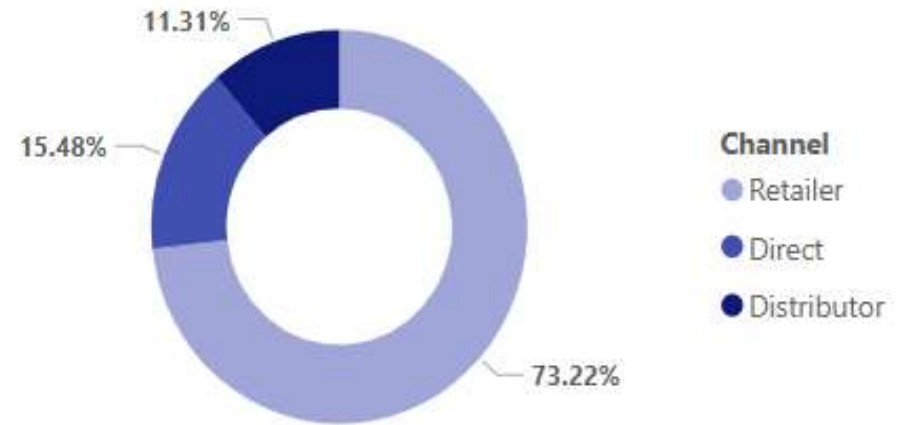
Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution?

Query

```
With cte1 as (  
  select  
    c.channel as channel,  
    round(sum(s.sold_quantity*g.gross_price)/1000000,2) as gross_sales_mln  
  from fact_sales_monthly s  
  Join fact_gross_price g  
  on s.product_code = g.product_code  
  Join dim_customer c  
  on s.customer_code = c.customer_code  
  where s.fiscal_year = 2021  
  group by c.channel  
)  
SELECT  
  channel,  
  gross_sales_mln,  
  concat(round(100*gross_sales_mln/sum(gross_sales_mln) over (),2),"%") as percentage  
FROM cte1  
order by percentage desc;
```

Output

channel	gross_sales_mln	percentage
Retailer	1924.17	73.22%
Direct	406.69	15.48%
Distributor	297.18	11.31%



Retailer contributed the most to gross sales with **73.22%**. While **Distributor** had the lowest contribution at **11.31%**.

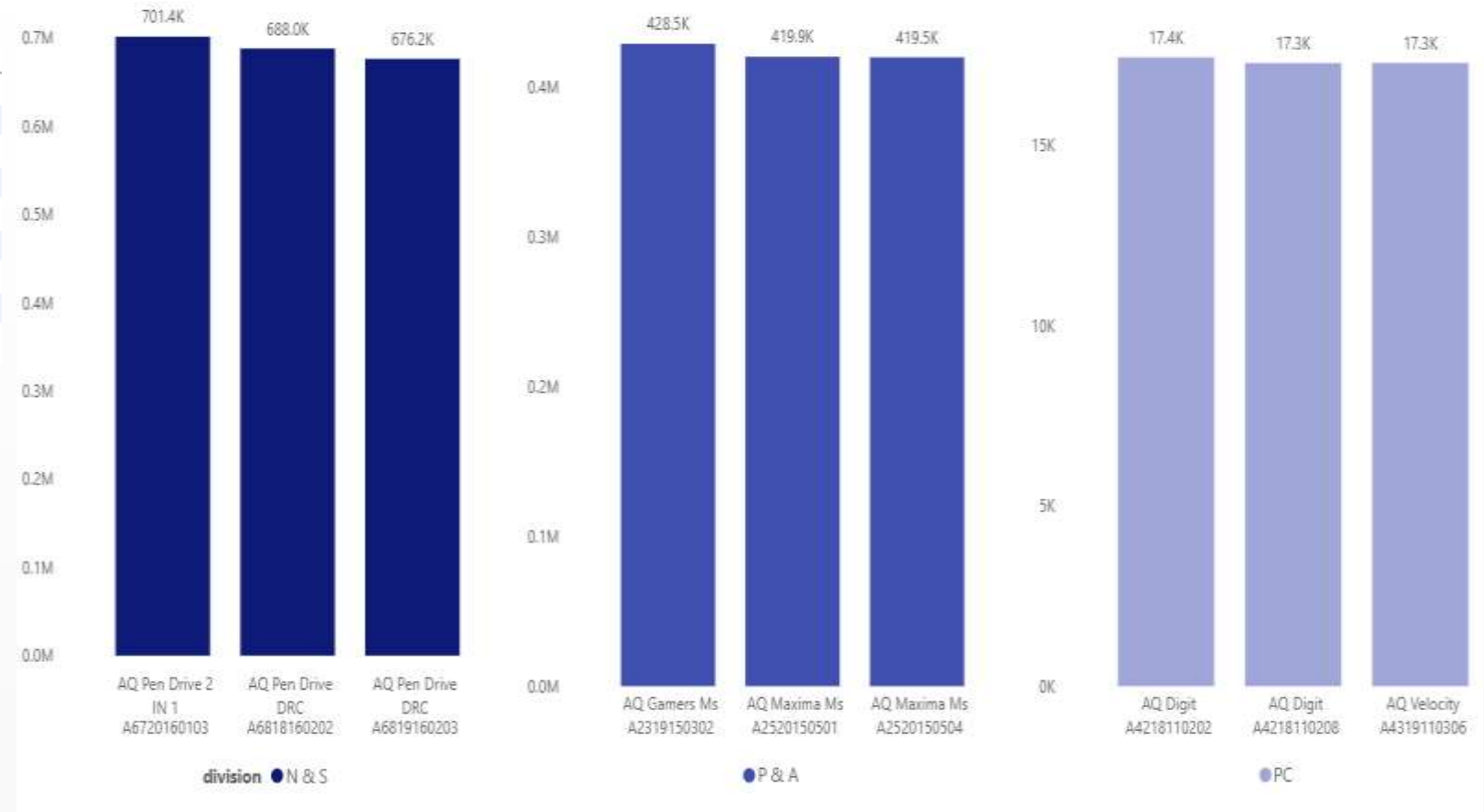
Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021?

Query

```
with cte1 as (  
  select  
    p.division,  
    p.product_code,  
    p.product,  
    sum(s.sold_quantity) as total_sold_quantity  
  from fact_sales_monthly s  
  join dim_product p  
  on s.product_code = p.product_code  
  where s.fiscal_year = 2021  
  group by p.division, p.product_code, p.product  
  order by total_sold_quantity desc  
,  
cte2 as (  
  select  
    *,  
    rank() over(partition by division order by total_sold_quantity desc) as rnk  
  from cte1  
)  
select * from cte2 where rnk<=3
```

Output

division	product_code	product	total_sold_quantity	rnk
N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
N & S	A6818160202	AQ Pen Drive DRC	688003	2
N & S	A6819160203	AQ Pen Drive DRC	676245	3
P & A	A2319150302	AQ Gamers Ms	428498	1
P & A	A2520150501	AQ Maxima Ms	419865	2
P & A	A2520150504	AQ Maxima Ms	419471	3
PC	A4218110202	AQ Digit	17434	1
PC	A4319110306	AQ Velocity	17280	2
PC	A4218110208	AQ Digit	17275	3



In each division the top 3 products have a highest total sold quantity in F.Y.2021.

THANK YOU

